Akoizo Original Practice - Form 1

Division B | Experimental Design | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) Which is most operational?

Unit: The experiment and the prompt packet.
Worked-case data: A prompt supplies a ramp, object, ruler, and timer. Turn “study motion” into a testable question.

A. Time in seconds between two fixed marks

B. How good the motion is

C. Whether the object seems fast

D. Interesting behavior

Answer: _____

2. (2 points) A fixed +0.5 s offset is…

A. A new independent variable always

B. No error

C. Systematic bias

D. Only random scatter

Answer: _____

3. (2 points) Which is a direct qualitative observation?

Unit: Materials, reproducible procedures, and qualitative data.
Worked-case data: A trial has no timer reading because the observer missed the endpoint. What should the table and procedure record?

A. Friction definitely increased

B. The hypothesis is true

C. The cause was gravity alone

D. The object wobbled

Answer: _____

4. (2 points) A missing trial should be fabricated to complete the table?

Unit: Practice, timing, teamwork, and final review.
Worked-case data: A team finishes data collection with little time left and no graph or analysis. Suggest a targeted practice change.

A. Only if software predicts it

B. No

C. Yes

D. Only if the hypothesis is clear

Answer: _____

5. (2 points) A source of error explanation should include…

Unit: Claims, evidence, and reasoning for trends and outliers.
Worked-case data: At levels 1,2,3, means are 12,16,20 units, with SD 2 at each level. Write a bounded trend CER.

A. Only a list of tools

B. Only an apology

C. How it changes the response

D. Only the word “human”

Answer: _____

6. (4 points) Write a directional ramp hypothesis.

Unit: Problem, hypothesis, variables, and experimentation.
Worked-case data: All low-level trials occur early and all high-level trials occur late while the instrument drifts upward.

Answer: _____
_____
_____

7. (6 points) Solve this case and explain your reasoning: 25 cm^2 towel: Absorbed masses: 2, 3, 4 g. 50 cm^2 towel: Absorbed masses: 5, 6, 7 g. Controls: Same towel brand, soak time, and draining method.

Answer: _____
_____
_____

8. (4 points) Distinguish SD and standard error.

Unit: Quantitative data, graphs, and statistics.
Worked-case data: For response values 8, 10, 12, calculate mean, range, and sample SD.

Answer: _____
_____
_____

9. (4 points) Why use several levels?

Unit: Problem, hypothesis, variables, and experimentation.
Worked-case data: All low-level trials occur early and all high-level trials occur late while the instrument drifts upward.

Answer: _____
_____
_____

10. (2 points) Define “Mean” in the context of Experimental Design.

Answer: _____
_____
_____

11. (4 points) Why can precise measurements be inaccurate?

Unit: Quantitative data, graphs, and statistics.
Worked-case data: For response values 8, 10, 12, calculate mean, range, and sample SD.

Answer: _____
_____
_____

12. (6 points) Solve this case and explain your reasoning: A prompt supplies a ramp, object, ruler, and timer. Turn “study motion” into a testable question.

Answer: _____
_____
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13. (6 points) Solve this case and explain your reasoning: Every timer reading is 0.5 s too high. Would ten more repeats eliminate the bias?

Answer: _____
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14. (2 points) Define “Qualitative data” in the context of Experimental Design.

Answer: _____
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15. (2 points) Define “Dependent variable” in the context of Experimental Design.

Answer: _____
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16. (2 points) Define “Systematic bias” in the context of Experimental Design.

Answer: _____
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17. (2 points) Define “Residual” in the context of Experimental Design.

Answer: _____
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18. (2 points) Define “Confounding” in the context of Experimental Design.

Answer: _____
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